1/21/2026

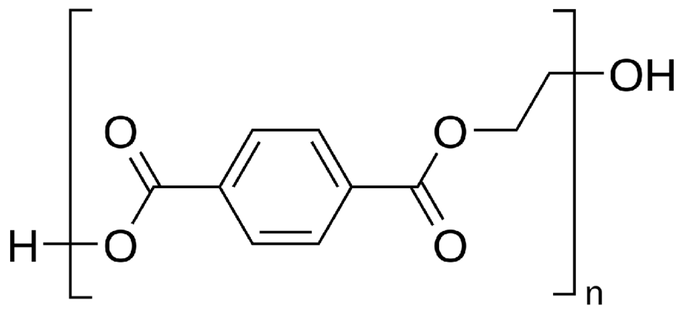
- Introduction to biological macromolecules
- Exercise 2: Structural visualization with nglview

Biological Macromolecules

- This lecture is intended to help you achieve the following learning objectives:
  - Recall the main types of biological macromolecules, identify their monomers, and describe their levels of structure.
  - Describe the types of forces that maintain the structure of biological macromolecules and stabilize their interactions with small organic molecules.
- It will be about how biological macromolecules
  - are heteropolymers
  - are folded
  - have noncovalent interactions with many drugs
- At the end of this mini-lecture, you should be able to answer the following questions:
  - What are biological macromolecules made of?
  - What does it mean for a biological macromolecule to be folded?
  - How do most drugs interact with their targets?

Biological macromolecules are heteropolymers

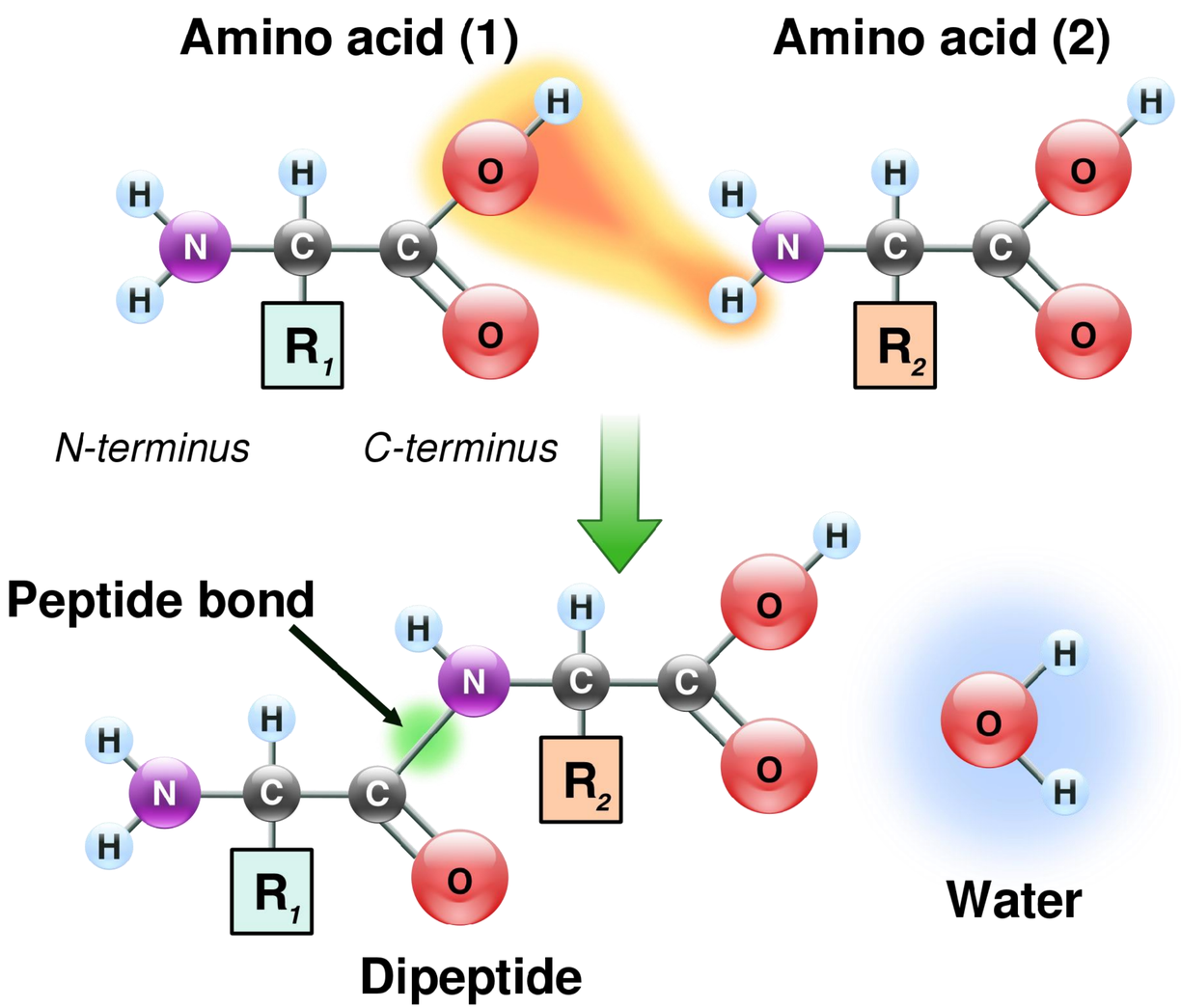
- Polymers - made of smaller building blocks - monomers - that are covalently joined together
  - Homopolymers - monomers repeat, e.g. in a plastic
  - Heteropolymers - monomers do not exactly repeat
- Different types of macromolecules are made of different types of building blocks



Polyethylene terephthalate, a homopolymer
https://commons.wikimedia.org/wiki/File:Polyethyleneterephthalate.svg

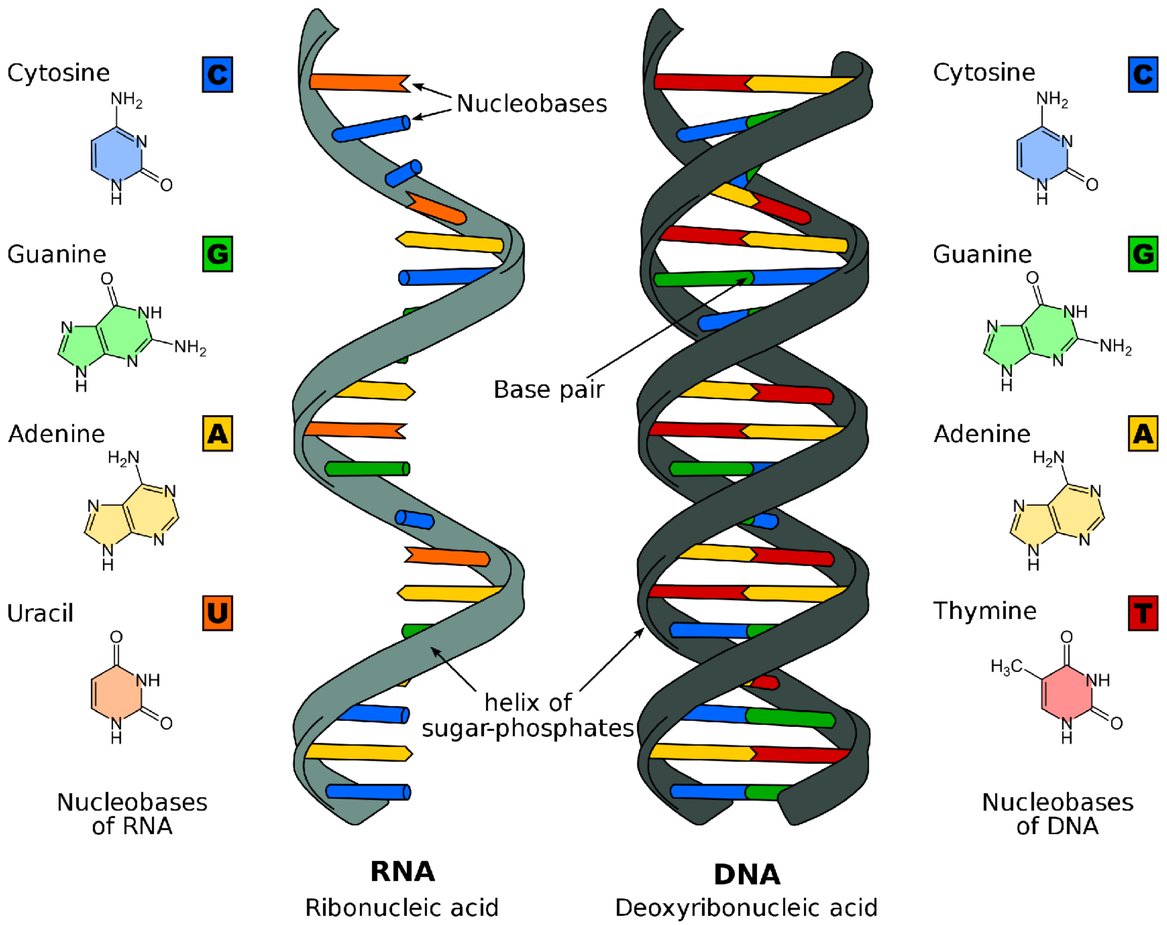
The monomers are small organic molecules

- Proteins are made of
  - 20 standard amino acids
  - linked by peptide bonds
  - modifications, e.g.
    - post-translational modification
    - disulfide bonds
    - cofactors and prosthetic groups



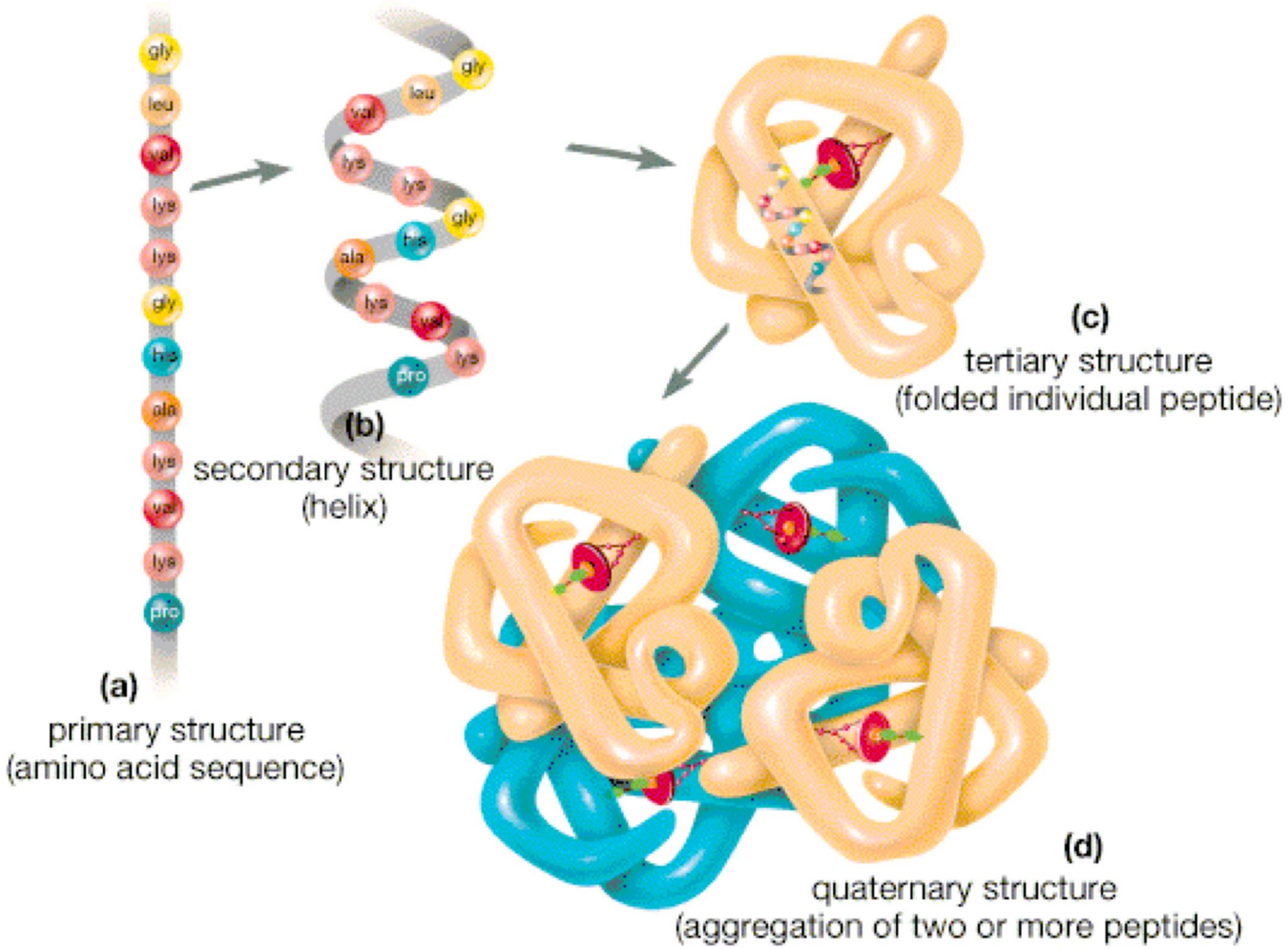
The monomers are small organic molecules

- DNA and RNA are made of nucleic acids
- DNA usually forms a double helix
- RNA is more flexible and can have complex structure
- Which can have catalytic activity? DNA or RNA?



https://en.wikipedia.org/wiki/Nucleic_acid#/media/File:Difference_DNA_RNA-EN.svg

There are four levels of protein structure



https://www.slideshare.net/SabahatAli9/protein-structure-levels

The primary structure is the monomer sequence

Primary Structure = sequence of amino acids

3-letter code

Lys-Thr-Tyr-Phe-Pro-His-Phe-Asp-Leu-Ser-His-**Gly**-...

1-letter code

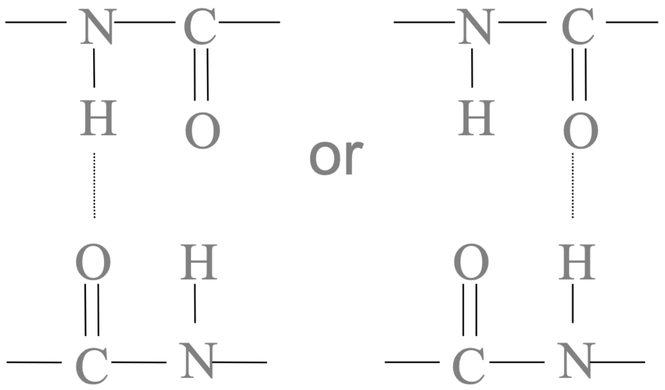
KTYFPHFDLSH**G**...

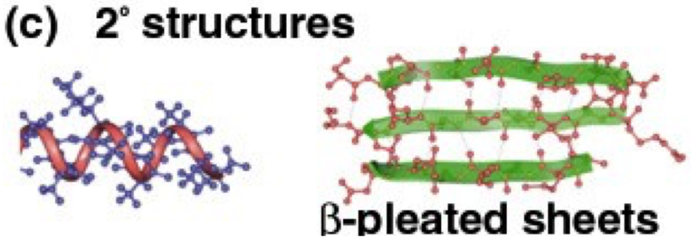
A ***chain*** is a sequence of amino acids connected by covalent peptide bonds. Chains are synthesized from the amino terminus, with new amino acids being added at the carboxy terminus.

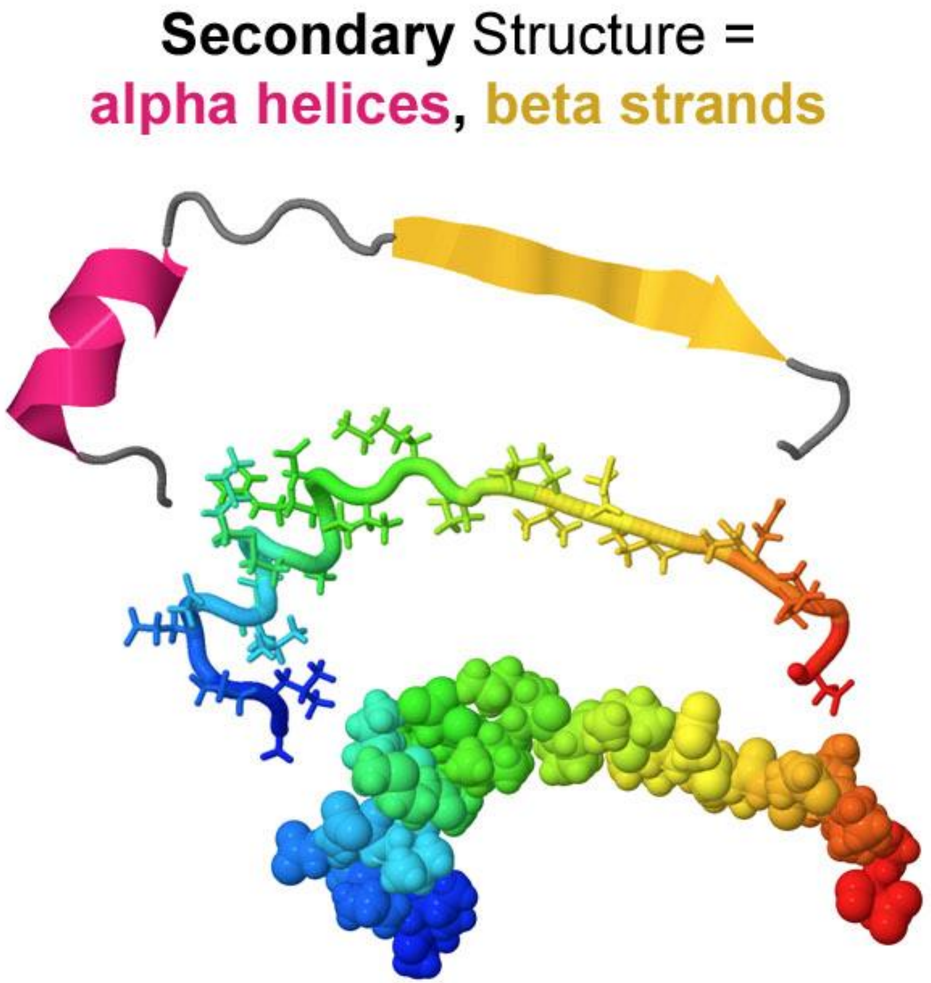
https://proteopedia.org/wiki/index.php/Four_levels_of_protein_structure

Many sequences have secondary structure

Folded structure due to hydrogen bonds between the amino and acid groups of amino acids





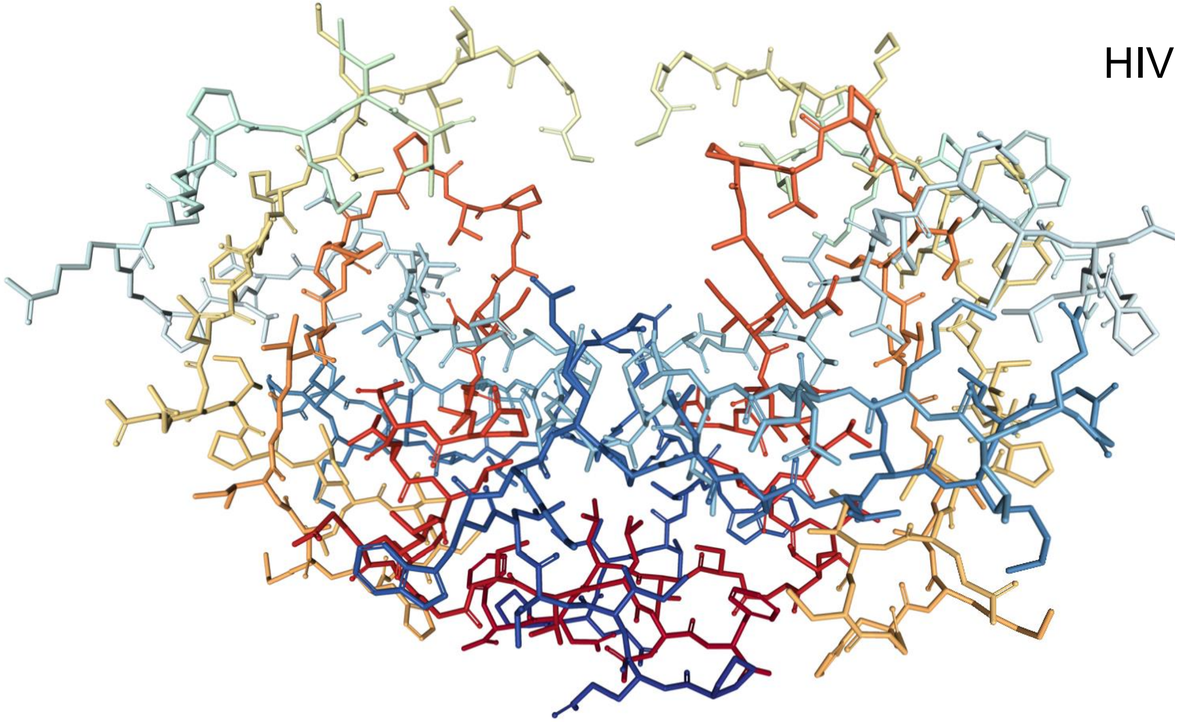


https://www.slideshare.net/SabahatAli9/protein-structure-levels

https://proteopedia.org/wiki/index.php/Four_levels_of_protein_structure

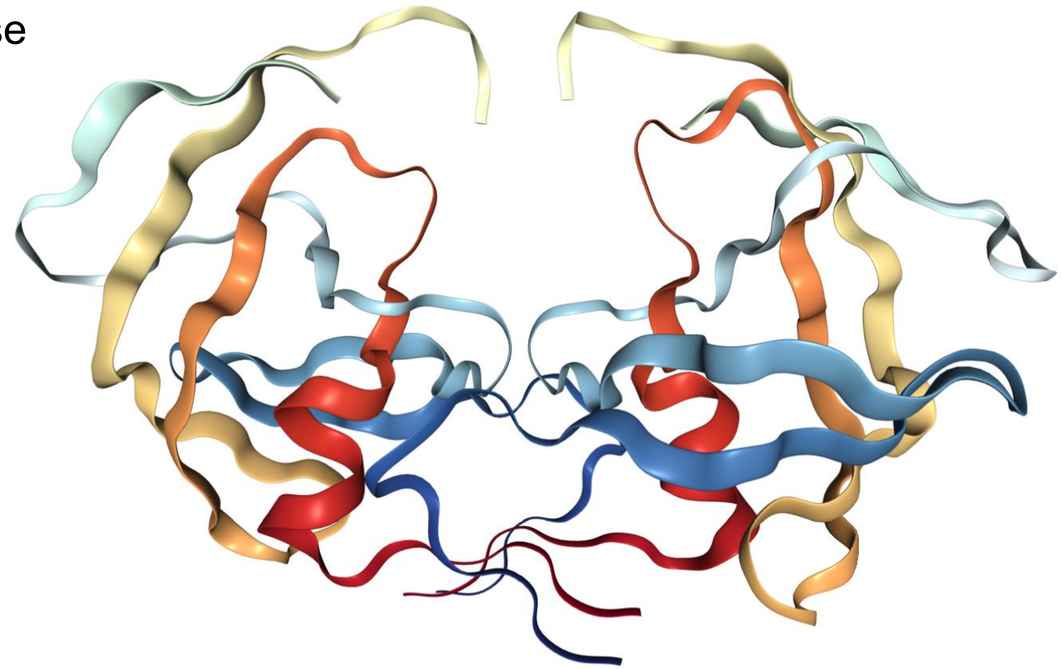
Proteins often form folded tertiary structures

HIV protease



“Licorice” view showing all heavy atoms

http://www.rcsb.org/3d-view/2HB2



“Ribbon” view showing backbone, emphasizing α helices and β sheets

“Folded” does not mean that they are completely rigid, but they are *fairly* well-defined.

Primary Structure = sequence of amino acids

3-letter code

Lys-Thr-Tyr-Phe-Pro-His-Phe-Asp-Leu-Ser-His-**Gly** ...

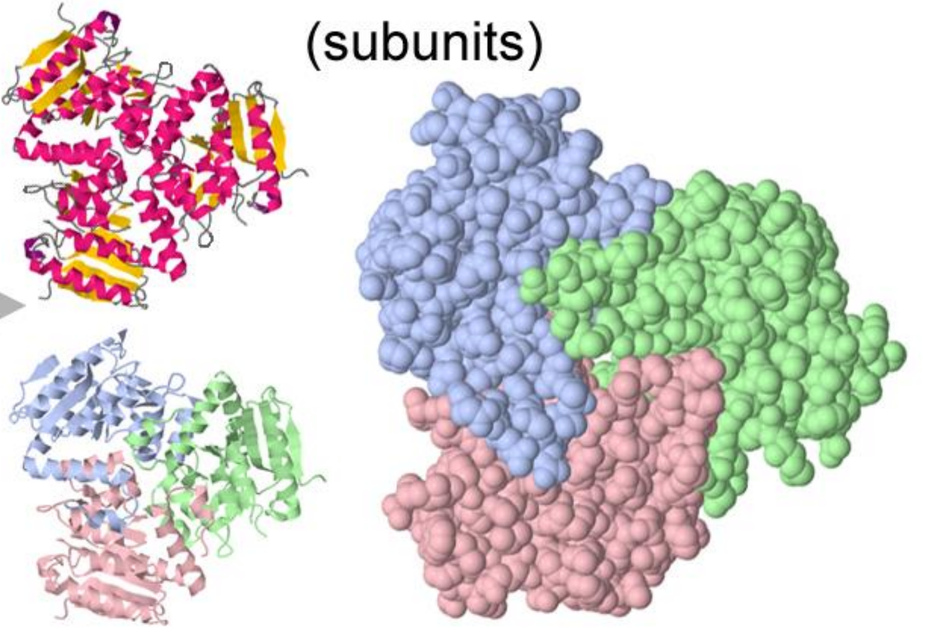
1-letter code

KTYFPHFDLSH**G**

Secondary Structure = **alpha helices**, **beta strands**

Tertiary Structure = fold helices and strands into domains

Quaternary Structure (Biological Units) = functional assemblies of chains (subunits)



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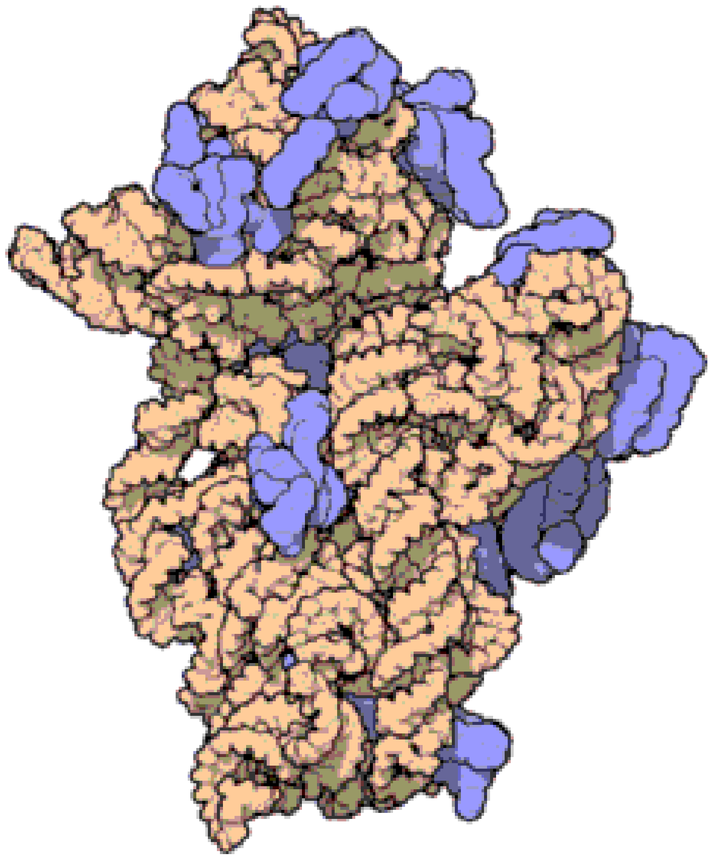
Complexes can include protein + nucleic acids



30S subunit from a bacterial ribosome,
which is made of both protein and RNA

Review Questions

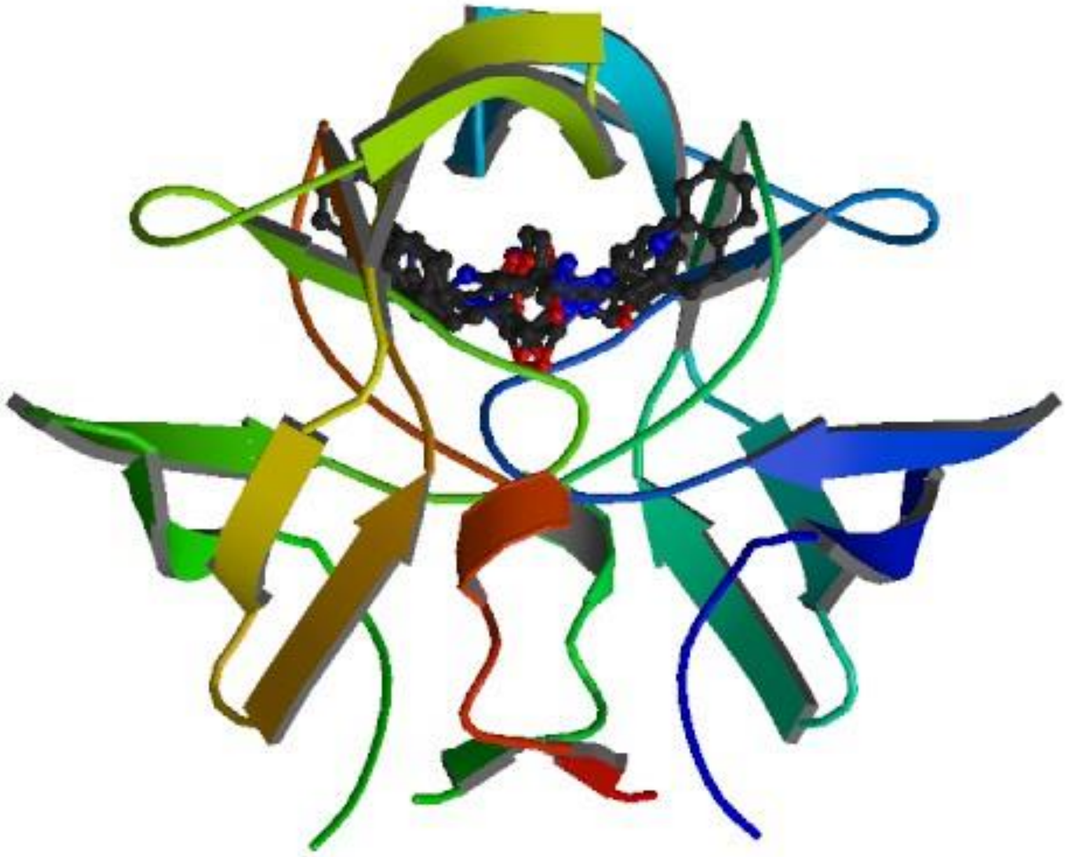
1. Primary
2. Secondary
3. Tertiary
4. Quaternary

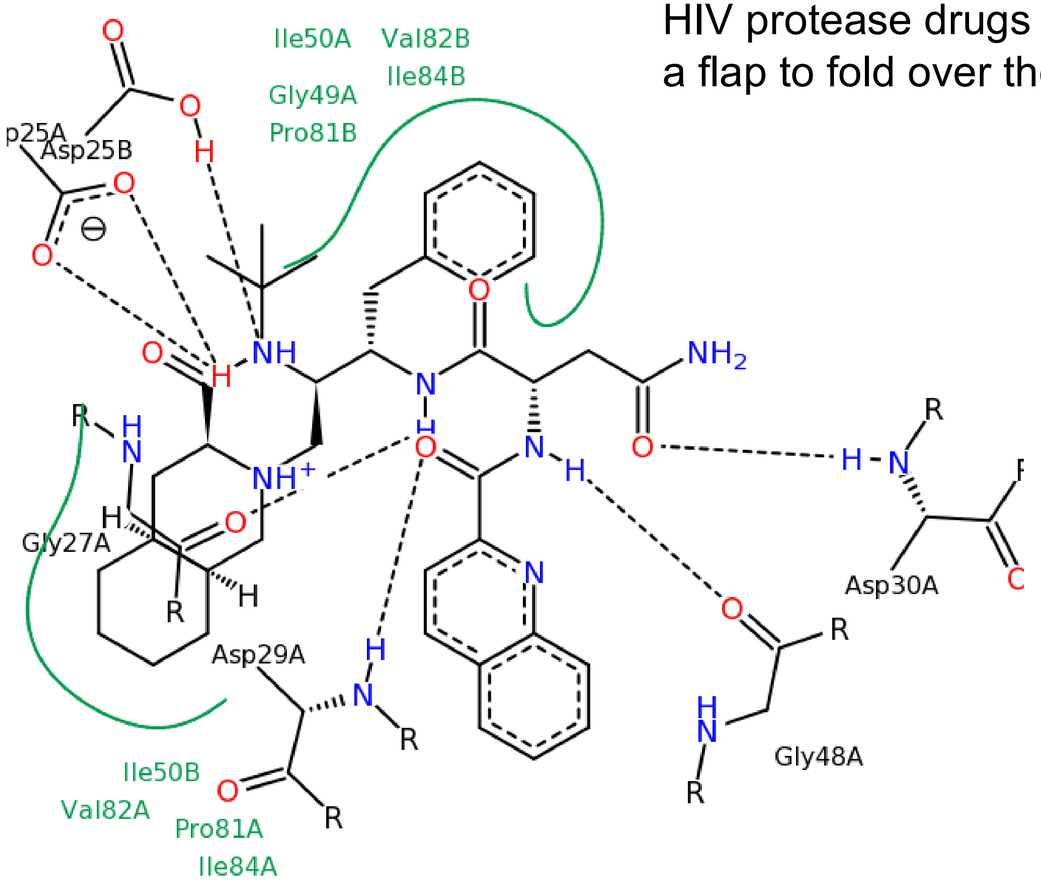
A. Which structure results exclusively from hydrogen bonding?
B. Which structure involves an association of two or more protein chains?
C. Which structure describes the linear sequence of amino acids?
D. Which structure depends upon interactions between the R groups of the amino acids?

https://www.slideshare.net/SabahatAli9/protein-structure-levels

Most drugs are small molecules that specifically interact with the folded structures

HIV protease with saquinavir



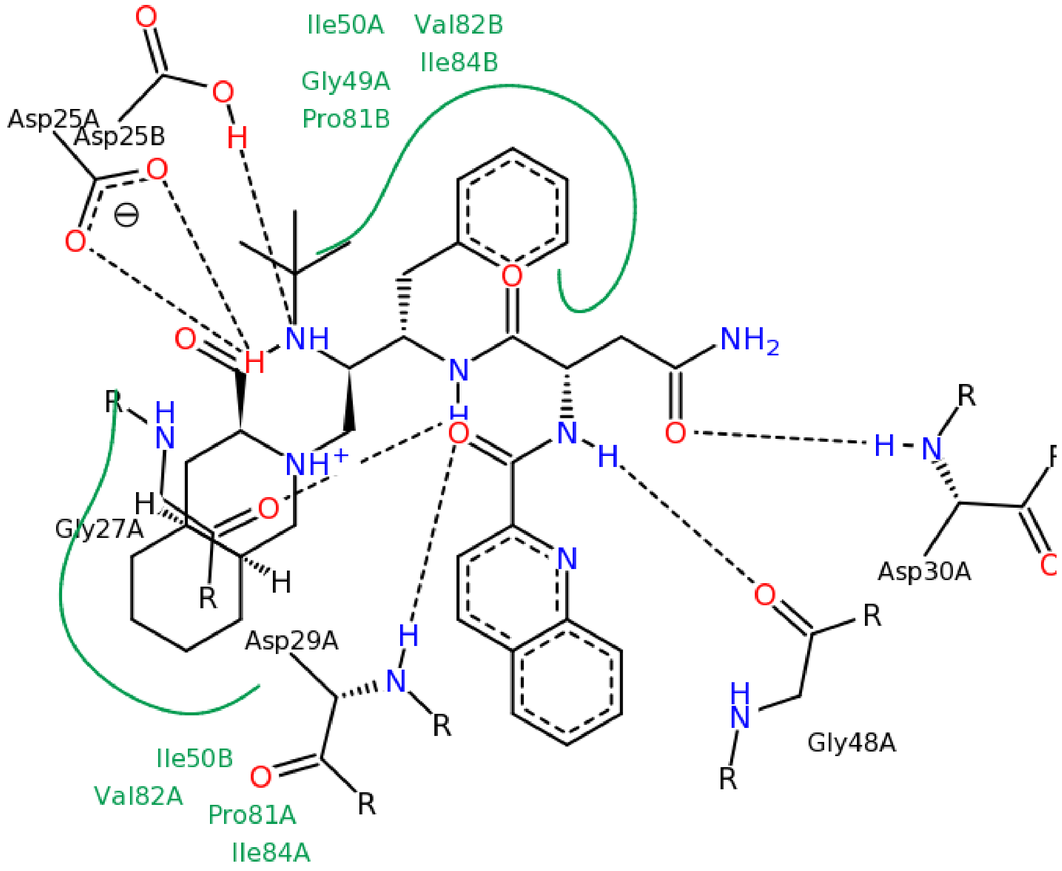


HIV protease drugs often cause a flap to fold over the active site

http://www.rcsb.org/pdb/101/motm_disscussed_entry.do?id=1hxb

Most drug-target interactions are noncovalent

- The interactions driving drug binding are primarily
  - steric - van der Waals. atoms like to be close but not *too* close.
  - electrostatic - like charges repel and opposite charges attract. H bonding often treated as electrostatic.
- Water can play an important role.
- Some drugs (like penicillin) bind to their targets covalently.



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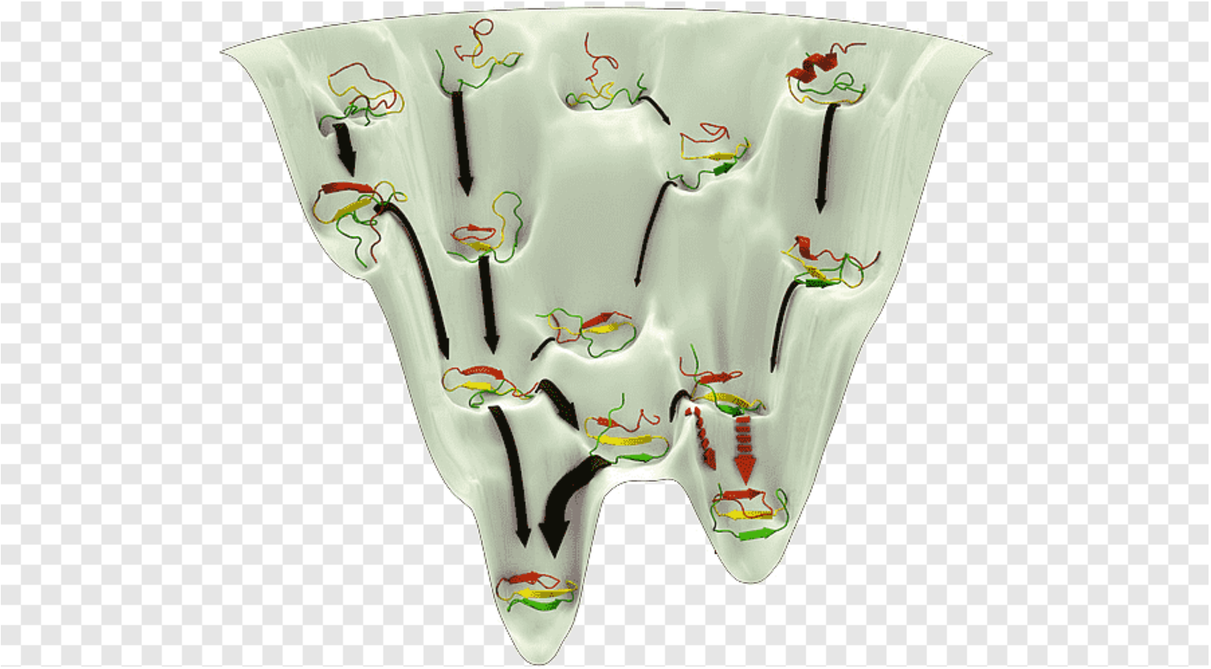
Folding and binding processes are described by energy landscapes

- Many configurations are possible
- There are many pathways between unfolded configurations
- Packing of hydrophobic side chains is a key driver of folding



https://www.pngegg.com/en/png-dyixm

Review Questions

- What are biological macromolecules made of?
- What does it mean for a biological macromolecule to be folded?
- How do most drugs interact with their targets?

Misc

- A beautiful short video on “A basic introduction to drugs, drug targets, and molecular interactions”: https://www.youtube.com/watch?v=u49k72rUdyc

Exercise 2: Structural visualization with nglview

Homework

- By Friday 1/23 at 8:30am, complete Exercise 1.
- By Wednesday 1/28 at 8:30am, complete Exercise 2.